

Materials Science Recommendation Report

Generated: 2026-06-03 23:10:53

User Request:

Photocatalytic TiO2 coating for water purification using UV light

Material Category:

Photocatalytic Coating

Target Application:

water purification using UV light

Composition:

Component	Ratio	Percent
Titanium dioxide (anatase)	0.850	85.0%
Silica binder	0.100	10.0%
Silver nanoparticles	0.050	5.0%

Category-Specific Parameters & Targets:

Substrate Type: Glass, ceramic, or polymer
Coating Thickness: 0.5-5 micrometers

Validation Plan:

Pollutant Degradation Efficiency: >80% degradation
Leaching Test: ICP-MS analysis

DISCLAIMER: All material parameters, compositions, and performance targets in this report are AI-generated planning defaults based on materials science knowledge. These parameters DO NOT demonstrate proven water-harvesting performance, potable-water safety, commercial readiness, or suitability for any specific application. All recommendations are CONDITIONAL upon rigorous laboratory validation, field testing, compliance with applicable water-quality standards, and consultation with qualified materials engineers and water-quality specialists. This report is for research and development guidance only.